

RVO-Guided RL Framework with Multi-Head Attention for Multi-UAV Pursuit and Dynamic Obstacle Avoidance

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Abstract—Multi-agent systems (MAS) have shown great advantages in complex tasks such as disaster response and UAV swarm control. However, problems such as the dynamic obstacle avoidance, real-time collaboration, and scalability still have not been addressed. To address these challenges, we propose the *Attention-based Reciprocal Velocity Policy Multi-Agent Deep Deterministic Policy Gradient* (ARVP-MADDPG) algorithm, a multi-agent reinforcement learning framework that integrates Multi-Head Attention, Prioritized Experience Replay (PER), and curriculum learning. Notably, we include Reciprocal Velocity Obstacles (RVO) strategy into the reward function to support the multi-UAV cooperation in the dynamic environment. First, UAV kinematics and detection models are constructed, and dynamic obstacles with linear motion patterns are simulated to create a realistic testbed. Subsequently, a reward function integrating Approaching Reward, Distance-Based Safety Reward, RVO-Based Safety Reward, 3-Stage Round-Up Reward and Completion Reward is designed to guide the UAV for tracking, encirclement, and capture tasks. The Multi-Head Attention boosts the cooperation efficiency in the centralized critic and the Prioritized Experience Replay (PER) improves the policy convergence. Finally, curriculum learning divides training into two stages: dynamic obstacle avoidance and task-specific encirclement, ensuring progressive performance improvement. Experimental results show that ARVP-MADDPG outperforms traditional methods and other reinforcement learning models in task completion rate, collision avoidance, and path planning efficiency, validating its robustness and applicability in simulation environments that reference real-world conditions. Comparative experiments demonstrate that ARVP-MADDPG increases the task completion rate by 64.39% and 63.30% compared to MADDPG and MATD3, while reducing the collision rate by 6.36% and 4.24%.

Index Terms—Multi-agent reinforcement learning, UAV swarm control, Dynamic obstacle avoidance

I. INTRODUCTION

Real-world tasks exhibit a high level of complexity, making it difficult for a single agent to accomplish these tasks independently. As an example, in domains such as environmental monitoring, disaster response, and security[1][2][3], the perception capabilities, computational resources, and action range of a single agent are often insufficient to efficiently handle complex tasks. Multi-Agent Systems (MAS) involve multiple agents collaborating to accomplish tasks, offering

significant advantages in various scenarios.[4] Their distributed control features provide flexibility and robustness in addressing complex challenges[5]. An increasing number of unmanned aerial vehicles (UAVs) are now equipped with embedded sensors and communication devices, drawing growing interest in UAV swarm control systems[6][7]. In multi-UAV swarm control, collaboration, and information sharing among agents are crucial.[8], [9] However, as the number of agents increases, the complexity of the system also increases, making it more difficult to maintain stability and improve the robustness of the control. The behavior of each agent is influenced not only by its own decisions, but also by the actions of others, creating a constantly changing environment during the learning process[10].

Collaboration and information sharing among agents become particularly crucial [11]. Traditional obstacle avoidance methods often depend on static environmental information, making them insufficient for dynamic obstacles. Although the Reciprocal Velocity Obstacle (RVO) algorithm shows good stability in multi-agent real-time navigation [12], it assumes that obstacles move uniformly linearly. This limitation makes it ineffective for environments with non-linear motion models [13]. ATT-MADDPG method models the dynamic joint policies of teammates [14], overcoming the limitations of traditional methods that hinder information exchange among agents. However, it still faces challenges such as slow training processes, reliance on static obstacle scenarios during training while real-world environments often feature dynamic obstacles, and limited generalization in multi-task scenarios. These issues complicate training and restrict its practical applicability in complex real-world tasks.

To enhance the performance of traditional dynamic obstacle avoidance algorithms in multi-UAV roundup tasks and address the challenges of target tracking and encirclement in multi-UAV swarm control, we propose the *Attention-based Reciprocal Velocity Policy Multi-Agent Deep Deterministic Policy Gradient* (ARVP-MADDPG) algorithm. The main contributions of this study are summarized as follows:

- Dynamic obstacle avoidance with DRL: A model was trained to handle dynamic obstacles in complex environments by integrating RVO reward-guided strategies with reinforcement learning, enhancing adaptability and robustness.
- Attention mechanism: Enhances the perception and information exchange capabilities of UAVs, thereby optimizing collaborative decision-making processes and enhancing the efficiency and robustness of Multi-Agent Systems in dynamic environments.
- Curriculum learning and prioritized experience replay: Progressive task design and optimized sample selection significantly accelerate the learning process, supported by experimental evidence that confirms the effectiveness of this methodology.

II. RELATED WORKS

A. Collaborative Multi-Agent Systems

In 1939, Lewin proposed a group cooperation model in his study of human cooperation (CoH), which posited that cooperation is contingent upon individual interactions and environmental influences. While this model illuminated fundamental principles of cooperation, it was inherently passive and lacked a clearly defined objective[15]. In practical contexts, individuals are required to perform their own tasks while concurrently collaborating to achieve collective goals. This concept closely aligns with the phased tasks involved in collaborative obstacle avoidance and coordination in multi-UAV systems [16][17]. Siddiqua et al. proposed an information-sharing matrix (ISM) framework to enhance the collaborative capabilities of robotic teams, ensuring high performance across varying team sizes and dynamic environments[18]. Similarly, Hu et al. introduced an innovative approach called CommFormer, which incorporates an Attention Unit to dynamically allocate weights to received information, significantly improving communication efficiency and coordination among agents. Their results show that the good communication between agents will benefit the team learning performance greatly [19].

B. Dynamic Obstacle Avoidance

D. Bareiss extended the LQR-Obstacles framework by incorporating velocity obstacles under linear differential constraints, enabling robots with dynamic characteristics to avoid static obstacles [20]. Traditional UAV obstacle avoidance architectures often separate task planning from collision-free motion planning. This simplifies problem handling but weakens integration between components, limiting adaptability and real-time performance. Task planning relies on computationally intensive deliberative methods for global path optimization, which depend on model accuracy. In contrast, obstacle avoidance uses reactive control for quick responses based on local perception, often resulting in local optima [21]. Hybrid control architectures are developed to integrate global optimization with rapid response, enhancing navigation performance. Nakhaeinia et al. proposed a three-layer hybrid control architecture comprising a deliberative layer for global path planning, a control execution layer, and

a reactive layer for dynamic adaptation and obstacle avoidance [22]. Similarly, He et al. introduced the HIPSO-MSOS algorithm, which improved path planning accuracy, convergence speed, stability, and robustness, though it remains limited in avoiding collisions with dynamic obstacles [23]. Wang et al. employed hybrid strategies in UAV formation trajectory planning to ensure spatial, temporal, and velocity coordination while maintaining relative positional relationships among UAVs [24]. Despite these advancements, increasing UAV numbers and environmental complexity continue to challenge traditional hybrid control methods in terms of computational complexity and scalability.

DRL-based obstacle avoidance strategies are emerged as a promising solution to address these limitations. In uncertain environments, RL has demonstrated robust decision-making abilities [25]. By continuously learning optimal strategies through interactions between agents and their environments, RL enables real-time decision-making without reliance on precise environmental models, making it particularly effective for multi-UAV collaborative obstacle avoidance and target tracking missions.

III. MODEL ESTABLISHMENT

A. UAV kinematics model

In a 2D plane from a bird's-eye view (BEV), the state of a UAV is defined by its position, velocity, and acceleration. The position components of UAV i at time step k are denoted as (x_i^k, y_i^k) . Its velocity components are characterized by $(v_{i,x}^k, v_{i,y}^k)$, while the acceleration components are represented as $(a_{i,x}^k, a_{i,y}^k)$.

The UAV kinematics model is described using a discrete-time system, considering the presence of certain sensor latency. The state-space equation is defined as follows:

$$\mathbf{x}_i^{k+1} = \mathbf{A}\mathbf{x}_i^k + \mathbf{B}\mathbf{u}_i^k \quad (1)$$

The dynamic modeling is decomposed into components along the x-axis and y-axis. The x-axis kinematic model is defined as follows:

$$\begin{bmatrix} x_i^{k+1} \\ v_{i,x}^{k+1} \\ v_{i,y}^{k+1} \\ a_{i,x}^{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 - \Delta t/T_d & \Delta t/T_d & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_i^k \\ v_{i,x}^k \\ v_{i,y}^k \\ a_{i,x}^k \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ b_{1x}\Delta t \end{bmatrix} u_{i,x}^k \quad (2)$$

The kinematic model of the y-axis is analogous to the x-axis model, with variables replaced by their respective components of the y-axis, including y_i^k , $v_{i,y}^k$, $v_{i,x}^k$, $a_{i,y}^k$ and $u_{i,y}^k$.

The yaw angle θ_i^k can be defined as $\theta_i^k = \text{atan}(v_{i,y}^k/v_{i,x}^k)$, where $p_i^k = [x_i^k, y_i^k]^T \in \mathbb{R}^2$, $v_i^k = [v_{i,x}^k, v_{i,y}^k]^T \in \mathbb{R}^2$, and $a_i^k = [a_{i,x}^k, a_{i,y}^k]^T \in \mathbb{R}^2$ represent the position, velocity, and acceleration. The sensor velocity $\mathbf{v}_{is}^k = [v_{isx}^k, v_{isy}^k]^T \in \mathbb{R}^2$ represents the velocity measurements accounting for sensor delay, where v_{isx}^k is the velocity along the x-axis and v_{isy}^k is the velocity along the y-axis. The sensor delay time is denoted by T_d , and specifically, for Global Navigation Satellite Systems (GNSS), T_d is 0.3s[26]. Additionally, Δt denotes the sampling interval, $u_{i,x}^k$ represents the x-axis throttle input signal for the

i -th UAV at time step k , and u_{iy}^k represents the y-axis throttle input signal for the i -th UAV at time step k . In this study, maximum velocity and acceleration are defined as v_m and a_m , respectively, with the UAV satisfying $|v^k| \leq v_m$ and $|a^k| \leq a_m$.

B. UAV Detection Model

In order to simulate the ability of UAVs to perceive environmental information, this paper designs a partially detectable UAV model based on LiDAR sensors.

LiDAR sensors return a distance value for each measurement direction, representing the distance to the nearest obstacle. Assuming the LiDAR sensor has n measurement directions, the measurement results can be expressed as $L_{\text{sensor}} = \{l_i \mid 1 \leq i \leq n\}$ [27]. The minimum safe radius for collision detection is defined as $R_{\text{collision}}$. When the minimum measured distance is less than or equal to $R_{\text{collision}}$, the UAV is considered to enter the collision area.

In the simulation, a collision is defined as the event where the distance between the geometric center of the UAV and the geometric center of an obstacle is less than the sum of the UAV's radius r_u and the collision radius, given by $R_{\text{collision}} = \lambda r_u$, where $\lambda \in [1.05, 1.2]$. If a UAV approaches an obstacle, a penalty is imposed that correlates with the distance to the obstacle; specifically, the closer the UAV is to the obstacle, the greater the penalty incurred. In instances of collision, a significant negative reward is assigned to incentivize the UAV to steer clear of collision-prone areas.

C. Obstacle Model

The obstacle model in the environment is constructed based on a constant linear velocity motion model. The environment contains n_0 obstacles, each represented as $O \in I_o = \{1, 2, \dots, n_0\}$. At time step k , the position of an obstacle is defined as the vector $\mathbf{p}_o^k = [x_o^k, y_o^k]^T \in \mathbb{R}^2$, and the velocity of the obstacle is defined as $\mathbf{v}_o^k = [v_{ox}^k, v_{oy}^k]^T \in \mathbb{R}^2$. Each obstacle is modeled as a circular area with a radius of r_o .

Obstacles move linearly at a constant velocity in the environment and will bounce back at the original speed upon touching the boundaries, with the direction satisfying the law of reflection. The update formula for the position and velocity vectors of the obstacles can be defined as follows:

$$\begin{bmatrix} \mathbf{p}_o^{k+1} \\ \mathbf{v}_o^{k+1} \end{bmatrix} = \begin{bmatrix} 1 & \Delta t I_2 \\ O & I_2 \end{bmatrix} \begin{bmatrix} \mathbf{p}_o^k \\ \mathbf{v}_o^k \end{bmatrix} \quad (3)$$

If the position of an obstacle exceeds the boundary of the workspace, it will bounce back, and the velocity component in the corresponding direction will be inverted. This model can be used for testing and validating dynamic obstacle avoidance algorithms, providing a reliable simulation foundation for dynamic obstacles and multi-UAV cooperative scenarios in this paper.

IV. ARVP-MADDPG

A. Multi-Head Attention in MADDPG

In this task, we extend the centralized critic Q_i from MADDPG, which has access to the global observations and actions of all agents, while the independent actor π_i only observes its own observation o_i . Drawing inspiration from ATT-MADDPG [14], we define a joint policy π_{-i} for other UAV agents. The joint action probability $\pi_{-i}(A_{-i}|s)$ characterizes the distribution over the action space A_{-i} of teammates under policy π_{-i} , and the action-value function $Q_i^{\pi_i}(s, a_i)$ is computed as follows:

$$\begin{aligned} Q_i^{\pi_i}(s, a_i) &= \mathbb{E}_{a_{-i} \sim \pi_{-i}} [Q_i^{\pi_i}(s, a_i, a_{-i})] \\ &= \sum_{a_{-i} \in A_{-i}} [\pi_{-i}(a_{-i}|s) Q_i^{\pi_i}(s, a_i, a_{-i})] \end{aligned} \quad (4)$$

To enhance collaboration, we introduce a Multi-head Attention mechanism [14][28] that dynamically adjusts the focus of the i -th agent on teammates' actions. A hidden vector $h_i(w_i)$, generated from the actions of teammates, encodes their behavior from the perspective of the i -th agent. The similarity between $h_i(w_i)$ and action-conditioned Q-values $Q_i^k(s, a_i|\vec{a}_{-i}; w_i)$ is computed using the dot product, resulting in attention weights $W_i(w_i)$, which are then normalized through the Softmax function as follows:

$$W_i^k(w_i) = \frac{\exp(h_i(w_i) Q_i^k(s, a_i|\vec{a}_{-i}; w_i))}{\sum_{k=1}^K \exp(h_i(w_i) Q_i^k(s, a_i|\vec{a}_{-i}; w_i))} \quad (5)$$

The attention weights are then used to compute a weighted sum of the action-conditioned Q-values, producing a contextual Q-value $Q_i^c(s, a_i, \vec{a}_{-i}; w_i)$ as follows:

$$Q_i^c(s, a_i, \vec{a}_{-i}; w_i) = \sum_{k=1}^K W_i^k(w_i) Q_i^k(s, a_i|\vec{a}_{-i}; w_i) \quad (6)$$

The contextual Q-value is finally passed through a fully-connected layer to produce the final Q-value. By focusing on teammates' critical actions and dynamically adjusting attention, this mechanism improves collaboration efficiency and decision-making quality, making agents' actions more intelligent and effective.

B. Design of states and actions

The state space of the UAV agent comprises four essential components: its own state, the positions of teammate UAVs, LiDAR data, and relative information regarding the target UAV. The overall dimensionality of the state space is 26. Specifically, the UAV's own state accounts for 4 dimensions, the positions of teammate UAVs contribute 2 dimensions, LiDAR data encompasses 16 dimensions, and relative information about the target UAV occupies 4 dimensions.

The simulation scenario is defined with a boundary width of W and a height of H . The state of the UAV, denoted as S_{uav_i} , encompasses normalized velocity and position information, which can be expressed as $S_{uav_i} = [x_i/W, y_i/H, v_{ixs}/v_m, v_{iys}/v_m]$, where x_i , y_i , v_{ixs} , and v_{iys} are obtained from the dynamic model equations, and v_m is the defined maximum flight speed of the UAV.

Based on the measurement accuracy of the simulated LiDAR sensor, the external state information perceived by the i -th UAV through LiDAR is normalized as $S_{laser_i} = [l_{1_i}, l_{2_i}, \dots, l_{16_i}] / L$ [27].

For the i -th UAV, the information about other UAVs S_{teamer_i} and the relative distance and bearing angle information of the target UAV S_{target_i} are defined as $S_{teamer_i} = [x_k/W, y_k/H, \dots, x_n/W, y_n/H], k \neq i$ and $S_{target_i} = [\frac{\|(x_e, y_e) - (x_i, y_i)\|_2}{\sqrt{W^2 + H^2}}, \arctan \frac{y_e - y_i}{x_e - x_i}]$, where x_e and y_e represent the horizontal and vertical coordinates of the target UAV. For the i -th UAV, the complete state information S_i is expressed as $S_i = [S_{uav_i}, S_{laser_i}, S_{teamer_i}, S_{target_i}]$.

In UAV kinematic modeling, given the control input $[u_{ix}^k, u_{iy}^k]^T$, the state-space equations along both the x -axis and y -axis can be solved to derive the state variables of UAVs along the x -axis as $[x_i^k, v_{ix}^k, v_{ix}^k, a_{ix}^k]^T$ and along the y -axis as $[y_i^k, v_{iy}^k, v_{iy}^k, a_{iy}^k]^T$. Furthermore, by manipulating the yaw angle and throttle inputs of the UAV, its state can be directly modified, thereby enabling control over its flight attitude and velocity. Consequently, the action space A_i of the i -th UAV, grounded in realistic physical parameters, can be defined as $A_i = [u_{ix}^k, u_{iy}^k]^T$.

C. Reward function

From a bird's-eye-view (BEV) in 2D, the UAV can navigate to the target point in this environment through the guidance of the comprehensive reward r_t . As shown in Figure 1, the comprehensive reward r_k at time $t = k$ consists of several components: Approaching Reward, Distance-Based Safety Reward, RVO-Based Safety Reward, 3-Stage Round-Up Reward and Completion Reward.

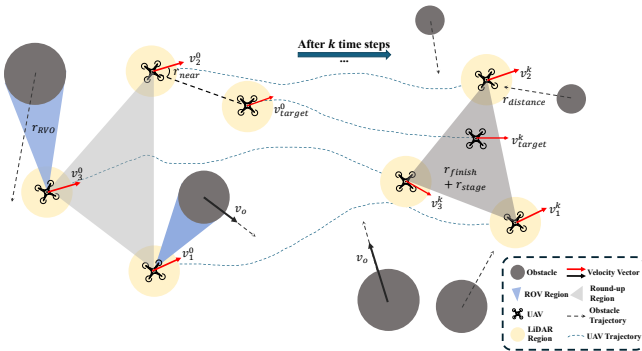


Fig. 1: Rewards in Round-up task

1) *Approaching Reward*: The calculation formula for the Approaching Reward is expressed as follows.

$$r_{near} = \left| \frac{v_i}{v_m} \right| \cdot \cos(\theta_i) \quad (7)$$

The purpose of the approaching target reward r_{near} is to encourage the UAV to gradually approach the target position p_d .

2) *Distance-Based Safety Reward*: The calculation formula for the Distance-Based Safety Reward is expressed as follows:

$$r_{distance} = \begin{cases} -10, & \text{if } \min(L_{sensor}) \leq R_{collision} \\ \left(\frac{\min(L_{sensor}) - \bar{L}_{sensor} - \varepsilon_{safe}}{L_{sensor}} \right), & \text{if } \min(L_{sensor}) > R_{collision} \end{cases} \quad (8)$$

The Distance-Based Safety Reward monitors the minimum distance between the UAV and surrounding obstacles or other UAVs using onboard sensors.

3) *RVO-Based Safety Reward*: The RVO-Based Safety Reward extends the traditional safety reward of the multi-agent system by incorporating the Reciprocal Velocity Obstacle (RVO) concept [29][30].

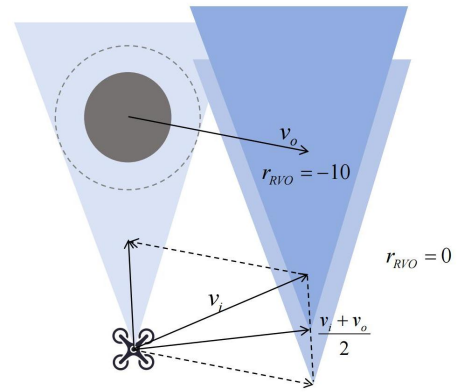


Fig. 2: RVO-Based Safety Reward

The general definition of the RVO is as follows:

$$RVO_O^U(v_O, v_U) = \{v'_U \mid 2v'_U - v_U \in RVO_O^U(v_O)\} \quad (9)$$

where $RVO_O^U(v_O, v_U)$ represents the set of velocities for the UAV agent U that are the average of its current velocity v_U and any velocity outside the RVO region relative to obstacle O . Geometrically, the RVO region is derived by translating the original RVO region such that its apex shifts to $(v_U + v_O)/2$, as illustrated in Figure 2.

The RVO-Based Safety Reward penalizes UAV agents when their velocity enters the RVO region of any obstacle, indicating a higher collision risk. The calculation formula for this reward is as follows:

$$r_{RVO} = \begin{cases} -10, & \text{if } v_U \in RVO_O^U(v_O, v_U) \\ 0, & \text{if } v_U \notin RVO_O^U(v_O, v_U) \end{cases} \quad (10)$$

4) *3-Stage Round-Up Reward*: The 3-Stage Round-Up Reward consists of three key components: Target Tracking Reward r_{track} , Encirclement Reward $r_{encircle}$, and Capture Reward $r_{capture}$, each corresponding to a specific phase of the task: tracking, encircling, and capturing the target. Based on the current positions of the i -th UAV, denoted as $p_i = (x_i, y_i)$, and the coordinates of the target tracking object represented by $p_e = (x_{target}, y_{target})$, the total area enclosed by the closed

polygon S_n , formed by n UAVs surrounding the target tracking object. The UAV set is defined as $U = \{p_1, p_2, p_3\}$, and the target set is defined as $E = \{p_e\}$.

To measure the tightness of the encirclement of the target by the agents, define the encirclement tightness index ρ as:

$$\rho = \frac{S_{U_1, U_2, U_3}}{\sum_{i=0}^{n-1} S_{U_i, U_{i+1}, E}} \quad (11)$$

where $S_{U_i, U_{i+1}, E}$ represents the area of the triangle formed by the i -th UAV and the $(i+1)$ -th UAV with the target. Specifically, for convenience, define $i=0$ and $i=n$ both to represent the n -th UAV. $S_{U_i, U_{i+1} \sim U_n}$ represents the area of the polygon formed by all n UAVs. When $\rho > 1$, p_e lies outside the triangle; when $\rho = 1$, p_e is located on the boundary of the triangle; and when $\rho < 1$, p_e resides within the triangle.

In stage 1 the conditions can be expressed as follows: (1) The target has not been effectively encircled by the UAVs, indicated by $\rho > 1$. (2) The cumulative distance between all UAVs and the target exceeds the established encircle distance threshold d_{limit} . (3) The distance from any individual UAV to the target is greater than the capture distance threshold $d_{capture}$.

If these conditions are met, the target tracking reward for the UAVs r_{track} , is defined by the equation:

$$r_{track} = -\frac{\sum_{i=1}^n d_{U_i E}}{\max_{i \in [1, n]} d_{U_i E}} \quad (12)$$

When $\rho > 1$, and the cumulative distance between all UAVs and the target is less than the established encircle distance threshold d_{limit} , the encirclement reward $r_{encircle}$ is defined as follows:

$$r_{encircle} = -\frac{1}{n} \ln \left(\sum_{i=0}^{n-1} S_{U_i, U_{i+1}, E} - S_{U_i, U_{i+1} \sim U_n} + \delta_{corr} \right) \quad (13)$$

Where δ_{corr} is a correction bias, with $\delta_{corr} \in (0.5, 1.7)$ in this experimental environment, used to smooth the reward by adjusting the sensitivity of the logarithmic function to input changes.

When $\rho \leq 1$, and the distance from any individual UAV to the target is greater than the capture distance threshold $d_{capture}$, the capture reward $r_{capture}$ is defined as follows:

$$r_{capture} = \exp \left(\frac{\sum_{p \in U} \|p - p_e\|_2^{(t-1)} - \sum_{p \in U} \|p - p_e\|_2^{(t)}}{n \cdot v_m} \right) \quad (14)$$

In conclusion, the 3-Stage Round-Up Reward provides a systematic evaluation of performance across the stages of tracking, encirclement, and capture, thereby ensuring the effective implementation of the encirclement strategy. The reward is formulated as $r_{stage} = r_{track} + r_{encircle} + r_{capture}$.

5) *Completion Reward*: In addition, upon completion of the three subtasks of tracking and encirclement, UAVs can receive a completion reward and denoted r_{finish} , if certain conditions are met. The conditions and the reward relationship are defined as follows:

$$r_{finish} = \begin{cases} 10, & \text{if } \sum_{i=0}^{n-1} S_{U_i, U_{i+1}, E} = S_{U_i, U_{i+1} \sim U_n} \\ & \text{and } \max(d_{U_i, E}) < d_{capture}, \\ 0, & \text{otherwise.} \end{cases} \quad (15)$$

The final comprehensive reward for each UAV, r_t is as follows:

$$r_t = \mu_1 r_{near} + \mu_2 r_{distance} + \mu_3 r_{RVO} + \mu_4 r_{stage} + \mu_5 r_{finish}, \quad (16)$$

where μ_i , with $i \in [1, 5]$, denotes the coefficient associated with each reward component.

V. MODEL TRAINING

A. Prioritized Experience Replay

In this paper, we introduce a Prioritized Experience Replay (PER) mechanism as an enhancement to the traditional uniform experience replay. The Critic network computes the temporal-difference (TD) error based on the current policy and state during training, thereby dynamically adjusting the priority of experience replay. Define i_j as the j -th sampled transition experience in a batch, N_b is the size of sampling in replay buffer and M_b is the maximum size of the replay buffer. The sampled batch data can be expressed as:

$$batch = \left\{ i_j \mid i_j \sim Discrete \left(\left\{ P(i) = \frac{p_i^\alpha}{\sum_k p_k^\alpha} \right\}_{i=1}^{N_b} \right) \right\}_{j=1}^{M_b} \quad (17)$$

where $p_i^\alpha = (|\delta_i| + \varepsilon)^\alpha$. α is a hyperparameter that controls the extent to which priority influences the sampling probability, δ_i is the TD-error, ε is a small value added to enhance the exploration of sampling probabilities and β is a hyperparameter that controls the degree of importance sampling bias correction. The importance sampling weight for each sampled experience is calculated as $w_i = \left(\frac{P(i)}{P(i)_{min}} \right)^{-\beta}$, where $P(i)_{min} = \min(P(i))$ is employed to ensure numerical stability.

B. Curriculum Learning

The training process is designed as a two-stage learning framework. In the first stage, the UAV is trained to acquire a dynamic obstacle avoidance strategy, while in the second stage, the UAV learns to execute the encirclement task.

Specifically, during the first stage, the three-stages round-up reward r_{stage} is disabled, and the weight of the safety reward r_{safe} is increased. This adjustment ensures that the UAV focuses exclusively on mastering fundamental dynamic obstacle avoidance behaviors. During this stage, the UAVs learn to navigate within the task boundary, avoid collisions with obstacles, and maintain proximity to the target.

In the second stage, leveraging the dynamic obstacle avoidance capabilities developed in the first stage, the 3-stages round-up reward r_{stage} is introduced. By incorporating stage-specific rewards, the model is guided to explore strategies for completing the target encirclement task. This approach enables the agent to learn how to achieve the task objectives while maintaining a high priority on obstacle avoidance.

VI. EXPERIMENT

The simulation experiment was conducted using NVIDIA GeForce RTX 2070 SUPER, Python 3.9, PyTorch 2.0.0, and CUDA 11.8, with the built-in model based on the Gym library. For this task, a virtual enclosed environment was constructed, where four circular obstacles with random velocities were randomly generated. The environment includes four agents: three UAVs modeled as quadrotor UAVs and one target represented as a red dot. The UAVs are assigned the task of tracking the target while avoiding dynamic obstacles, whereas the target UAV aims to evade both the UAVs and dynamic obstacles.

A. Parameter Description

The initial positions of Unmanned Aerial Vehicles (UAVs), obstacles, and the target are random values on a 2D plane, while their velocities are random numbers sampled from a uniform distribution within a specific interval. In the two-stage network hyperparameter configuration, the learning rate of the optimizer for the actor network is set to $l_\alpha = 0.00001$, while the learning rate of the optimizer for the critic network is set to $l_\beta = 0.001$. The discount factor is $\gamma = 0.99$ and the soft update rate $\tau = 0.005$. During the first stage, the number of training episodes is set to 5000. The coefficients associated with each reward component are defined as $\mu_1 = 0.9$, $\mu_2 = 0.2$, $\mu_3 = 0.0$, $\mu_4 = 10$, and $\mu_5 = 0.1$. In the second stage, the number of training episodes remains at 5000, with adjustments made to the coefficients such that they become $\mu_1 = 0.9$, $\mu_2 = 0.2$, $\mu_3 = 0.4$, $\mu_4 = 10$, and $\mu_5 = 0.1$. In the comparative experiments, the parameter settings of MADDPG and MATD3 are kept consistent with those of ARVP-MADDPG. The detailed simulation parameters are provided in Table 1.

B. Evaluation Metrics

To comprehensively assess the performance of the proposed model in the multi-UAV pursuit task, five metrics are designed to evaluate the completion rate, collision rate, path length, smoothness, and steady state error.

1) *Completion Rate*: The completion rate measures the percentage of successful task completions over the total number of experimental episodes.

2) *Collision Rate*: The collision rate quantifies the percentage of episodes in which UAVs collide with obstacles or other agents.

3) *Path Length*: The path length measures the total distance traveled by a UAV during the task. $Path\ Length = \sum_{t=1}^{T-1} \sqrt{(x_{t+1} - x_t)^2 + (y_{t+1} - y_t)^2}$, where T represents the total number of time steps. The coordinates (x_t, y_t) denote the position of the UAV at time step t .

4) *Smoothness*: The smoothness measures the cumulative change in the heading angle of the UAV during the task. $Smoothness = \sum_{t=1}^{T-1} |\theta_{t+1} - \theta_t|$, where $\theta_t = \arctan 2(y_{t+1} - y_t, x_{t+1} - x_t)$ is the angle of heading of the UAV at the time step t ; T is the total number of time steps.

TABLE I: Simulation Parameters

Parameter	Value	Description
N_u	3	The default number of UAVs
N_o	4	The default number of obstacles
N_t	1	The default number of Target UAV
H	2	Simulation scenario boundary width(m)
W	2	Simulation scenario boundary height(m)
Δt	0.5	Simulation time step(s)
r_o	0.15 – 0.20	The radius of each obstacle(m)
r_u	0.1	The radius of each UAV(m)
v_o	0.02 – 0.05	The velocity of obstacles(m/s)
v_m	0.12	Maximum speed of UAV(m/s)
a_m	0.06	Maximum acceleration of UAV(m/s ²)
v_{tm}	0.03	Maximum speed of target(m/s)
a_{tm}	0.04	Maximum acceleration of target(m/s ²)
$d_{capture}$	0.3	Capture distance threshold(m)
d_{limit}	0.55	Encircle distance threshold(m)
T_m	5000	The maximum step in each episode
L	0.2	Detection range of sensor(m)
M_b	1000000	The maximum size of the replay buffer
N_b	256	The size of sampling in replay buffer
α	0.6	Priority sensitivity to sampling probability
β	0.4	Importance sampling parameter
ε	10^{-5}	The constant to enhance exploration
L_{ep}	110	The total number of steps in an episode
K_h	4	The number head in K-head Module
l_α	0.00001	Learning rate of action network
l_β	0.001	Learning rate of critic network
γ, τ	0.99, 0.005	ARVP-MADDPG parameters

5) *Steady-State Error*: The steady-state error quantifies the final distance between the UAV and the target at the end of the task. $Steady - State\ Error = \frac{1}{N} \sum_{i=1}^N \|p_i - p_{target}\|$, where N is the number of UAVs. p_i is the final position of the i -th UAV and p_{target} is the target position.

C. Comparison with State-of-the-Art Models

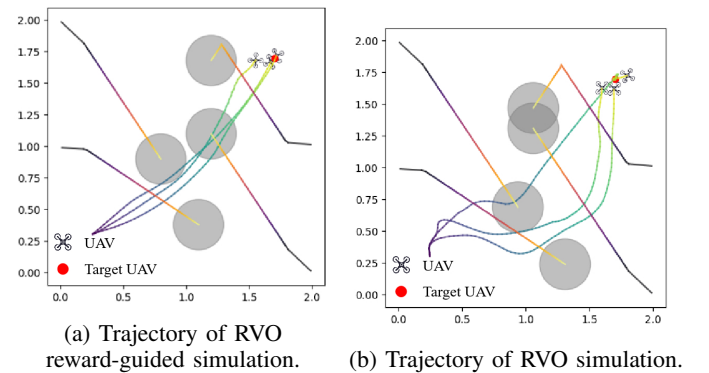


Fig. 3: Trajectory comparison between RVO algorithm and RVO reward-guided simulation.

The comparative experiment evaluates the performance of ARVP-MADDPG against three existing approaches in the multi-UAV rounding-up task: MADDPG, MATD3, and RVO.

TABLE II: Experimental Results of Multi-UAV Rounding-up Task

Model/Metric	Completion Rate (%)	Collision Rate (%)	Path Length (m)	Smoothness (rad)	Steady-state Error (m)
ARVP-MADDPG(ours)	96.84	7.28	2.15	8.33	0.096
MADDPG	32.45	13.64	2.66	16.92	0.103
RVO	0	18.80	2.37	8.83	0.118
MATD3	33.54	11.52	2.46	13.93	0.164

The RVO strategy relies on the current velocity of obstacles for linear obstacle avoidance but lacks predictive capabilities for future trajectories, often resulting in inefficient routes and reactive responses. In contrast, the proposed method integrates RVO into the reward function as a guiding mechanism, enabling optimized route learning through training. This approach is termed RVO reward-guided and demonstrates improved robustness and efficiency in complex scenarios. And see in Table 2.

As shown in Figure 3, simulation results highlight trajectory differences between conventional RVO and reinforcement learning-based strategies in multi-UAV rounding-up tasks. The reinforcement learning approach effectively optimizes path planning by incorporating RVO into the reward mechanism, enhancing task execution efficiency and robustness while making the planning process more goal-oriented.

Compared to the existing more general multi-agent reinforcement learning models, ARVP-MADDPG introduces targeted improvements that enable it to excel in complex dynamic environments. ARVP-MADDPG achieves the highest task completion rate (96.84%) and the lowest collision rate (7.28%), indicating its robust task execution and obstacle avoidance capabilities. Furthermore, it exhibits the shortest path length (2.15m) and the highest path smoothness (8.33rad), reflecting its predictive obstacle avoidance capabilities. Additionally, ARVP-MADDPG records the smallest steady-state error (0.096m), emphasizing its precision in reaching target positions. In contrast, models such as MATD3 and MADDPG demonstrate moderate performance but reveal shortcomings in critical metrics, such as task completion rate, smoothness, and steady-state error. These findings highlight the superior effectiveness and reliability of ARVP-MADDPG in addressing the challenges of multi-agent coordination and dynamic obstacle avoidance.

D. Ablation Experiments on Model Components

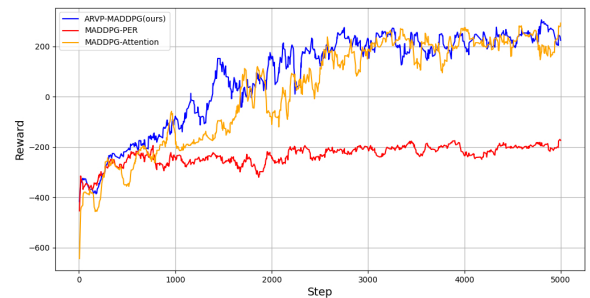
The ablation experiment evaluates the impact of individual components on the performance of APVP-MADDPG across curriculum learning stages using reward functions and five metrics: task completion rate, collision rate, total path length, path smoothness, and steady-state error.

The experimental results of the first stage are shown in Table 3. In the first stage, the minimum collision rate of the proposed model is 3.74%, and the minimum steady-state error approaching the target is 0.46m. This indicates that the proposed model can reach the target more quickly within the same number of episodes. Removing the Multi-Head Attention module increases the collision rate by 10.61%, demonstrating that Multi-Head Attention promotes information exchange and

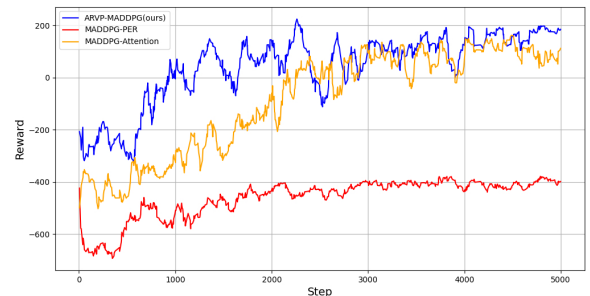
collaborative learning among drones. However, the total path length is only 0.49m, indicating that the drones adopt an overly conservative obstacle avoidance strategy, prioritizing immediate collision avoidance over efficient path planning, which affects overall performance in subsequent stages. Removing the RVO strategy from the reward function leads to a decline in obstacle avoidance performance, resulting in an increase in the collision rate by 6.46%.

In the second stage, the proposed ARVP-MADDPG model achieved a 96.04% task completion rate, significantly outperforming other models, and reduced the collision rate to 7.28% by leveraging a dynamic reward function and RVO strategy. The model also balanced path length, smoothness, and steady-state error, with prioritized experience replay (PER) accelerating policy convergence. These results demonstrate the model's efficiency and robustness in multi-UAV collaboration under complex dynamic environments.

E. Training Process Analysis



(a) The average rewards on stage 1



(b) The average rewards on stage 2

Fig. 4: The average rewards on different stages

The average rewards of 15 independent experiments are shown in Figure 4. From the learning content of the 2-stage

TABLE III: Experimental Results of the First Stage

Model/Metric	Completion Rate (%)	Collision Rate (%)	Path Length (m)	Smoothness (rad)	Steady-state Error (m)
ARVP-MADDPG(ours)	0	3.74	2.02	12.95	0.46
MADDPG-PER-RVO	0	14.35	0.49	2.96	1.80
MADDPG-ATT-RVO	0	6.84	2.14	4.85	1.09
MADDPG-ATT-PER	0	10.2	1.69	7.91	0.80

TABLE IV: Experimental Results of the Second Stage

Model/Metric	Completion Rate (%)	Collision Rate (%)	Path Length (m)	Smoothness (rad)	Steady-state Error (m)
ARVP-MADDPG(ours)	96.04	7.28	1.93	9.25	0.085
MADDPG-PER-RVO	64.45	9.49	2.30	19.37	0.076
MADDPG-ATT-RVO	90.53	13.36	1.66	7.065	0.16
MADDPG-ATT-PER	94.57	8.28	1.96	9.97	0.13

curriculum learning, it is evident that the trends of the reward values in the two stages are generally consistent. Our model demonstrates superiority over the ablation models in terms of both convergence speed and final reward value. As show in Table 4.

Since the removal of RVO fundamentally alters the reward distribution and precludes a meaningful comparison, our analysis focuses on two configurations: the elimination of Multi-Head Attention and the implementation of prioritized experience replay. Compared to models that do not incorporate attention mechanisms, our model demonstrates superior performance in terms of rewards. Similarly, the integration of PER significantly accelerates convergence, underscoring its effectiveness in training.

VII. CONCLUSION

This paper introduces the *Attention-based Reciprocal Velocity Policy Multi-Agent Deep Deterministic Policy Gradient* (ARVP-MADDPG) algorithm for dynamic obstacle avoidance and collaborative tasks in multi-UAV systems. By integrating UAV kinematics, a multi-component reward function, Multi-Head Attention, Prioritized Experience Replay (PER), and curriculum learning, ARVP-MADDPG achieves higher task completion rates, lower collision rates, efficient path planning, and reduced steady-state errors in complex environments.

Ablation studies confirm the importance of of Multi-Head Attention, PER, and RVO-based rewards in enhancing collaboration efficiency, accelerating policy convergence, and optimizing obstacle avoidance strategies. The integration of these components ensures robust decision-making and efficient learning in multi-agent scenarios.

Future work will focus on deploying the algorithm to real-world autonomous vehicles and UAV systems to validate its performance in practical scenarios.

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